

Machine Learning: The No-Nonsense (But Fun) Guide — Day 19

Hierarchical Clustering: The "Family Tree" of Data

1. Story & Intuition

Yesterday we learned K-Means, which is fast and simple, but it has one important limitation—you must specify the number of clusters (K) before training.

But what if you don't know how many clusters actually exist?

Hierarchical Clustering solves this problem by building a tree of relationships among all data points. Instead of deciding the number of clusters beforehand, you first build the hierarchy and then decide where to cut the tree.

Imagine discovering a new island with unknown animals. Initially, every animal is considered unique. Gradually, similar animals are grouped together until eventually all animals belong to one large family tree.

Hierarchical Clustering works in exactly the same way.

K-Means: "Tell me the number of clusters first."

Hierarchical Clustering: "Let's build the relationships first, then decide."

2. What is Hierarchical Clustering?

Hierarchical Clustering groups similar observations by building a hierarchy of clusters.

There are two approaches:

Agglomerative (Bottom-Up)	Divisive (Top-Down)
Starts with every point as its own cluster	Starts with one large cluster
Continuously merges the closest clusters	Continuously splits clusters
Most commonly used	Rarely used
Efficient and practical	Computationally expensive

In practice, Agglomerative Clustering is almost always used.

3. Working of Agglomerative Clustering

Consider six data points:

Point	X	Y
A	1	1
B	2	1
C	5	1
D	6	2
E	1	4
F	2	5

Step 1

Treat every point as an individual cluster.

[A] [B] [C] [D] [E] [F]

Step 2

Compute the distance between every pair of points.

The closest pair is merged into a single cluster.

For example:

- Merge A and B
 - New cluster becomes [A,B]
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Step 3

Recalculate distances between the new cluster and the remaining clusters.

How this distance is calculated depends on the Linkage Method.

Step 4

Again merge the closest clusters.

Repeat until only one cluster remains.

4. Linkage Methods

The linkage method determines how the distance between two clusters is measured.

4.1 Single Linkage (Nearest Neighbor)

Uses the minimum distance between any two points belonging to different clusters.

Advantages

- Detects irregular shapes
- Finds elongated clusters

Disadvantages

- Sensitive to noise
- Suffers from the chaining effect

Best Used For

Irregular and chain-like clusters.

4.2 Complete Linkage (Farthest Neighbor)

Uses the maximum distance between two clusters.

Advantages

- Produces compact clusters
- Less affected by noise

Disadvantages

- Splits large natural clusters

Best Used For

Compact and well-separated clusters.

4.3 Average Linkage

Uses the average distance between all pairs of points belonging to two clusters.

Advantages

- Balanced approach
- More stable than single linkage

Disadvantages

- Slightly slower

Best Used For

General-purpose clustering.

4.4 Ward Linkage

Ward linkage merges the two clusters that produce the smallest increase in within-cluster variance.

Advantages

- Produces compact clusters
- Very popular
- Less sensitive to noise

Disadvantages

- Works best for spherical clusters
- Uses Euclidean distance

Best Used For

Most real-world clustering problems.

4.5 Centroid Linkage

Measures the distance between the centroids of two clusters.

Advantages

- Easy to understand

Disadvantages

- Can create inversions in the dendrogram
 - Less commonly used
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5. Complete Example

Initial Clusters

[A] [B] [C] [D] [E] [F]

Iteration 1

Merge:

A + B

New cluster:

[A,B]

Iteration 2

Merge:

E + F

New cluster:

[E,F]

Iteration 3

Merge:

C + D

New cluster:

[C,D]

Iteration 4

Merge:

[A,B] + [E,F]

Iteration 5

Merge everything into one cluster.

6. Dendrogram

A Dendrogram is a tree diagram showing how clusters merge.

It helps determine the optimal number of clusters by cutting the tree at a chosen height.

Interpretation

- Vertical axis → Merge distance
 - Horizontal lines → Merge operations
 - Cut the tree horizontally to obtain the desired number of clusters.
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7. Distance Metrics

Metric	Suitable For
Euclidean	Continuous numerical data
Manhattan	Grid-like data
Cosine	Text mining
Correlation	Shape similarity
Mahalanobis	Correlated features

8. Time Complexity

Operation	Complexity
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Distance Matrix	$O(n^2)$
Memory	$O(n^2)$
Single Linkage	$O(n^2)$
Complete Linkage	$O(n^2 \log n)$
Ward Linkage	$O(n^2)$
Divisive Clustering	$O(2^n)$

Hierarchical Clustering is not recommended for extremely large datasets.

9. When Should You Use Hierarchical Clustering?

Use Hierarchical Clustering When

- Dataset is small or medium-sized
- Number of clusters is unknown
- Relationships between observations are important
- Dendrogram visualization is required
- Taxonomy or hierarchy needs to be created

Avoid Hierarchical Clustering When

- Dataset is very large

- Fast clustering is required
 - Memory is limited
 - Number of clusters is already known
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10. Real-World Applications

Domain	Application
Biology	Phylogenetic trees
Marketing	Customer segmentation
Healthcare	Disease grouping
Finance	Portfolio clustering
Social Networks	Community detection
Document Analysis	Topic hierarchy
Image Processing	Image segmentation

11. Common Pitfalls

Problem	Solution
Chaining effect	Use Complete or Ward linkage
Different feature scales	Standardize data before clustering
Wrong linkage selection	Compare multiple linkage methods
Outliers	Remove outliers before clustering
Misinterpreting dendrogram	Validate with silhouette score

12. Interview Questions

Basic

- 1. What is Hierarchical Clustering?**
- 2. What is the difference between Agglomerative and Divisive Clustering?**
- 3. Explain Single, Complete, Average, and Ward Linkage.**
- 4. What is a Dendrogram?**

Intermediate

- 1. What is the chaining effect?**

2. Why is Ward linkage commonly preferred?
 3. How does the choice of distance metric affect clustering?
 4. How do you determine the optimal number of clusters?
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Advanced

1. Why is Hierarchical Clustering computationally expensive?
 2. How would you cluster one million data points?
 3. Explain Cophenetic Correlation.
 4. Compare Hierarchical Clustering with Minimum Spanning Trees.
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Happy learning !!!!!